ARTIFICIAL INTELLIGENCE(Q) – APRIL 2026

KECERDASAN TIRUAN(Q) – APRIL 2026

CONVOLUTIONAL NEURAL NETWORKS

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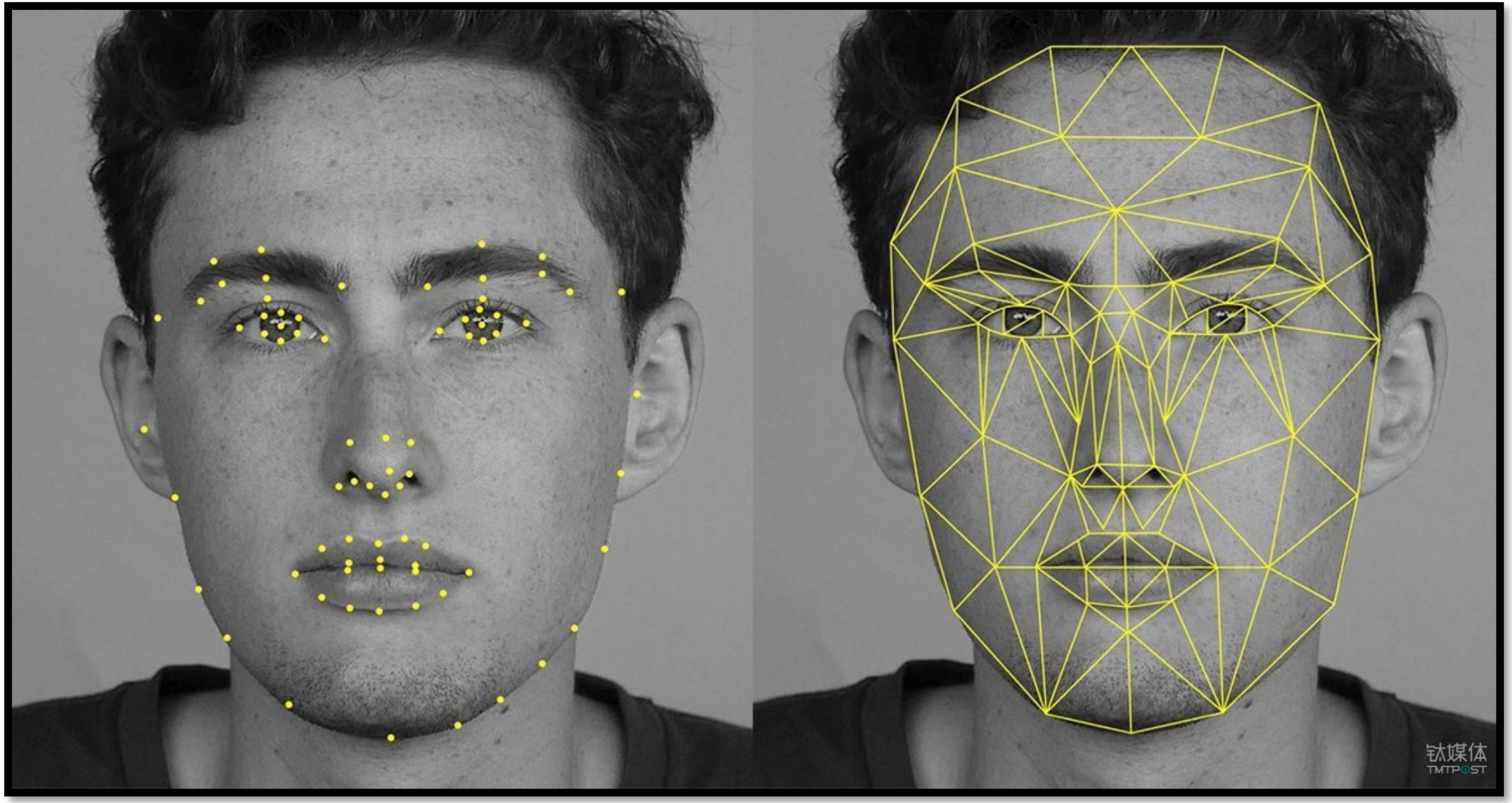
OUTLINES

- Introduction
- Computer Vision
- What computer learns?
- Visual Feature Learning
- Feature Extraction and Convolution
- Convolutional Neural Network
- Non-Linearity and Pooling
- End-to-End Code

Computer Vision

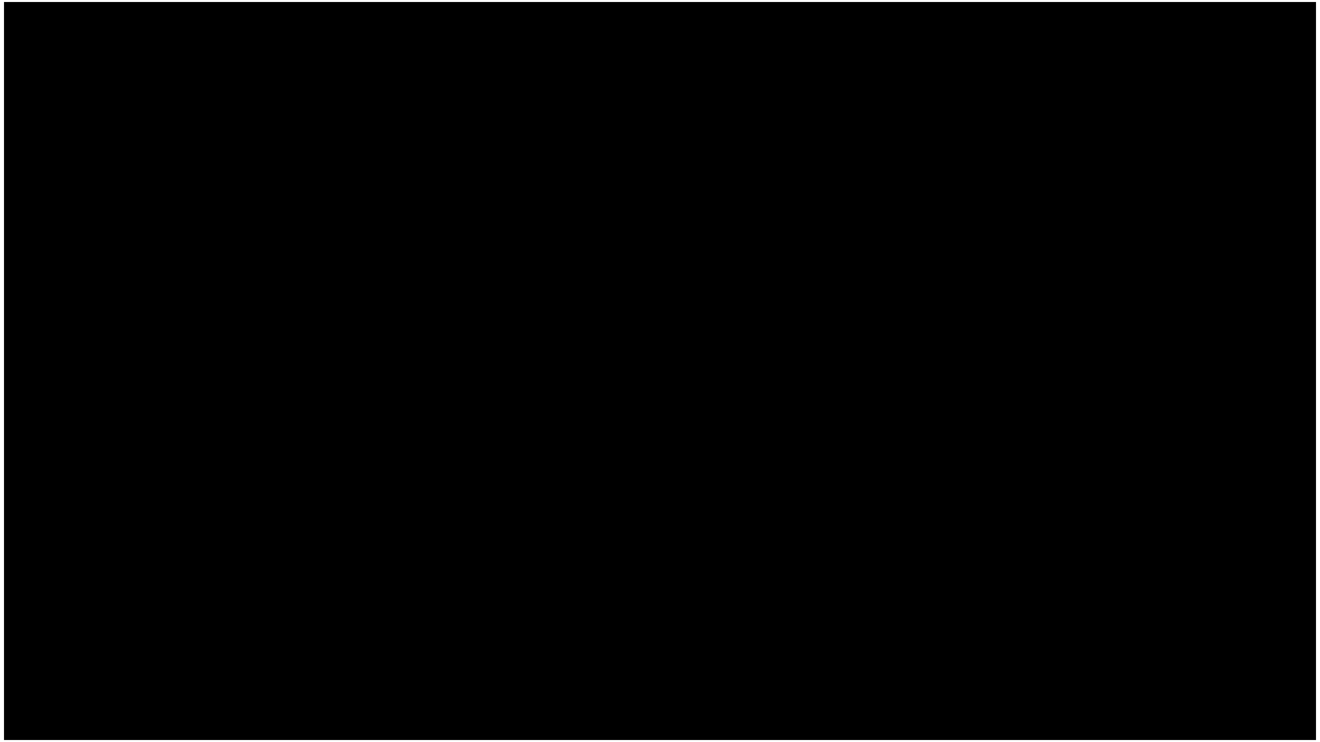
Face Detection & Recognition

https://rapidapi.com/blog/top-facial-recognition-apis/



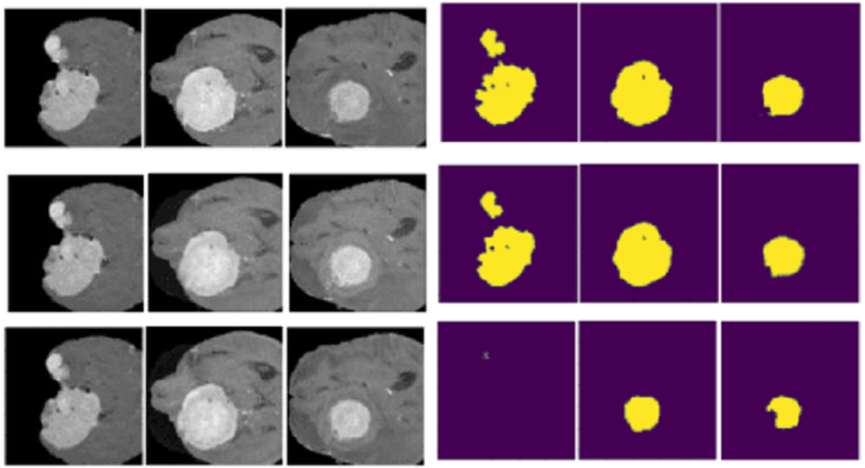
Self Driving Car

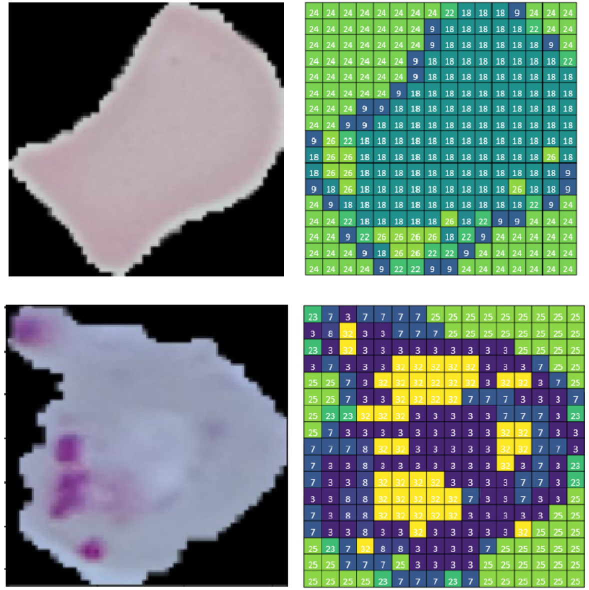
https://www.tesla.com/autopilot



Healthcare

Segmentasi Tumor Otak





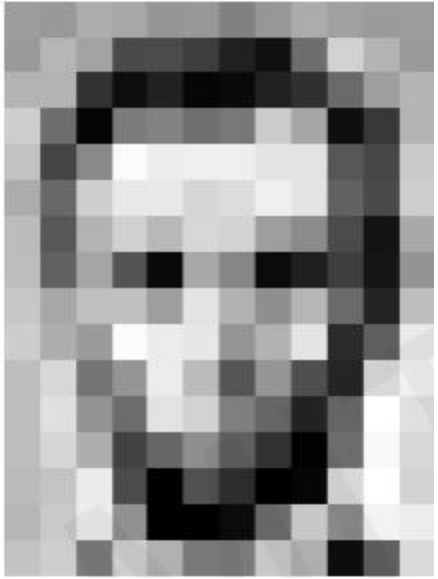
Sel darah yang terinfeksi

What Computer Learns?

Image

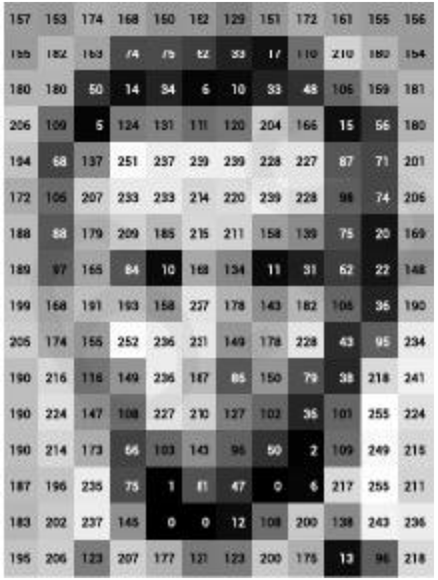
http://introtodeeplearning.com/

Seen Image



Computer Knows

157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

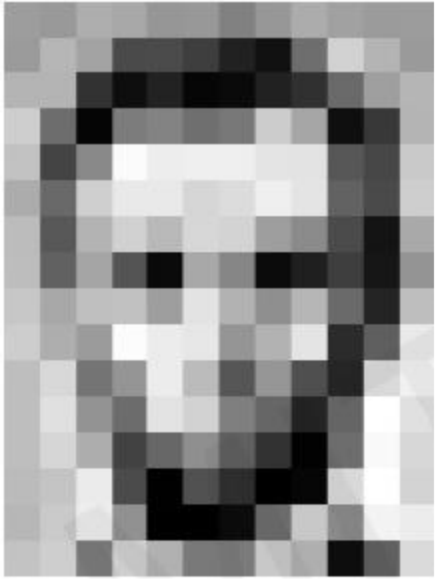


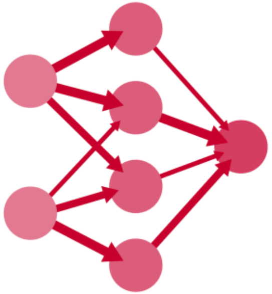
Connection

Task on Computer Vision…

http://introtodeeplearning.com/

Input Image





Feature Representation

157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

Classification

$$\begin{bmatrix} \text{Andi} \\ \text{Ani} \\ \text{Yanti} \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.1 \\ 0.1 \end{bmatrix}$$

Regression

Continuous value

What is the important?

http://introtodeeplearning.com/

Identifying main Feature on Image data

- Nose
- Eyes
- Lips

- Wheels
- Plate Number
- Car Headlight

- ✓ Door
- ✓ Window
- ✓ Chair

Manual Feature Extraction

http://introtodeeplearning.com/

Domain Knowledge → Define Feature → Detect Feature to classify

Viewpoint Variation

Illumination condition

Scale Variation

Deformation

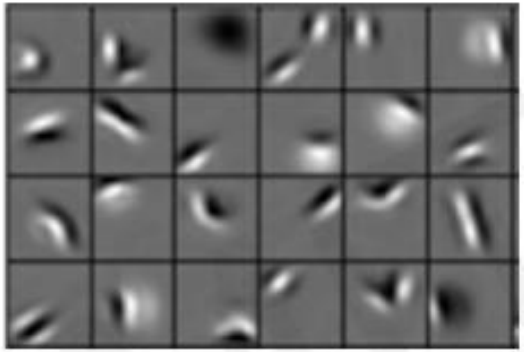
Occlusion

Background clutter

Intra-class variation

Learning Feature Representation

http://introtodeeplearning.com/

Low Level	Mid Level	High Level
		

- Edge
- Dark Spot

- Eye
- Ears
- Nose

✓ Facial Structure

Hand Engineering? Computationally expensive! a.k.a Annoying!

Can we learn hierarchy of features directly?

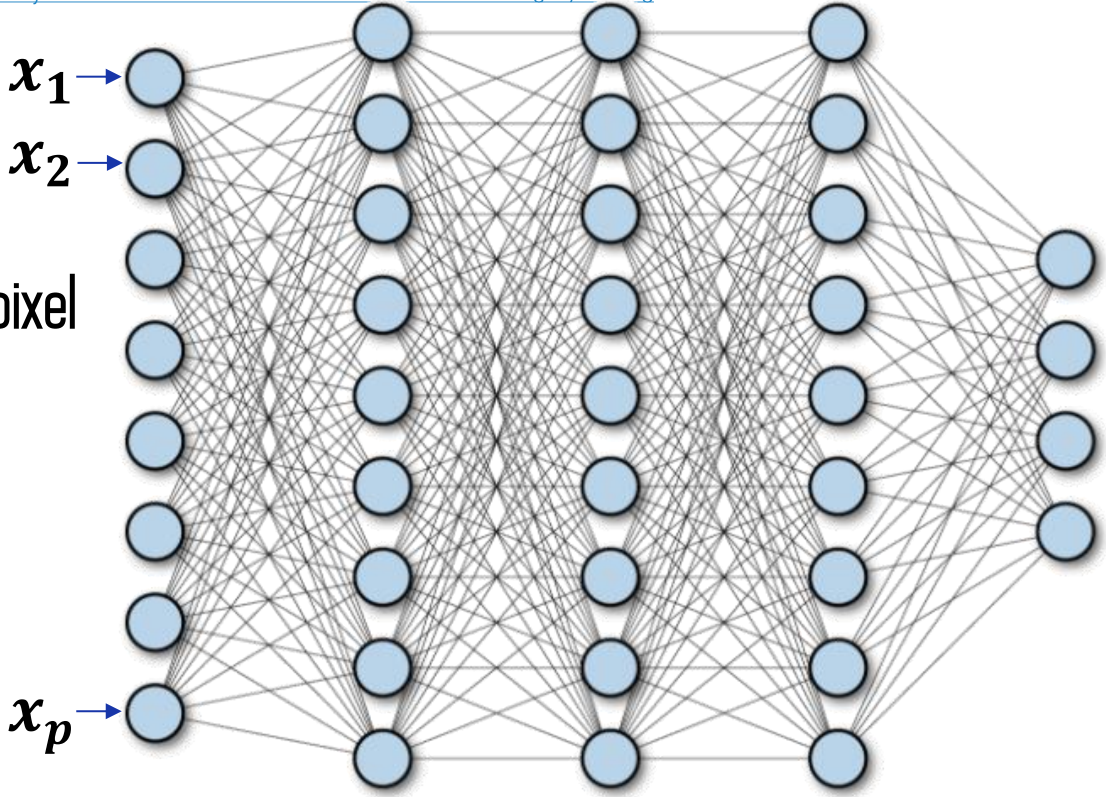
Learning Visual Features

Recall, Fully Connected Neural Networks

https://medium.com/swlh/fully-connected-vs-convolutional-neural-networks-813ca7bc6ee5

Input

- 2D image
- Vector of pixel values



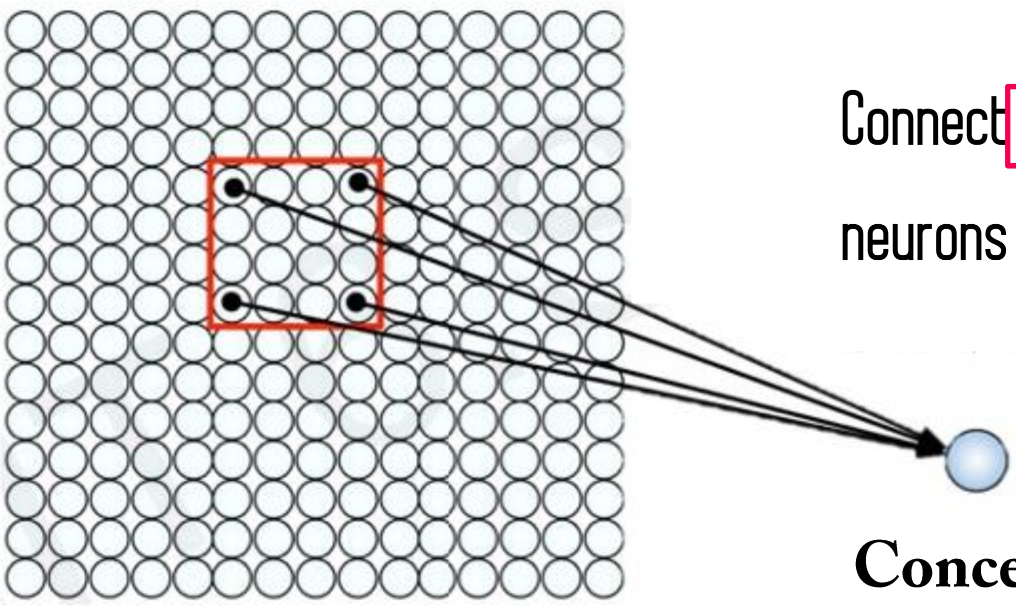
- Connect neuron in hideen layer to all neurons in input layer
- No spatial Information!
- And a bunch of parameters!

How we can get spatial feature in the input to inform the NNs?

Using Spatial Structure

Input

- 2D image
- Array of pixel values



Idea

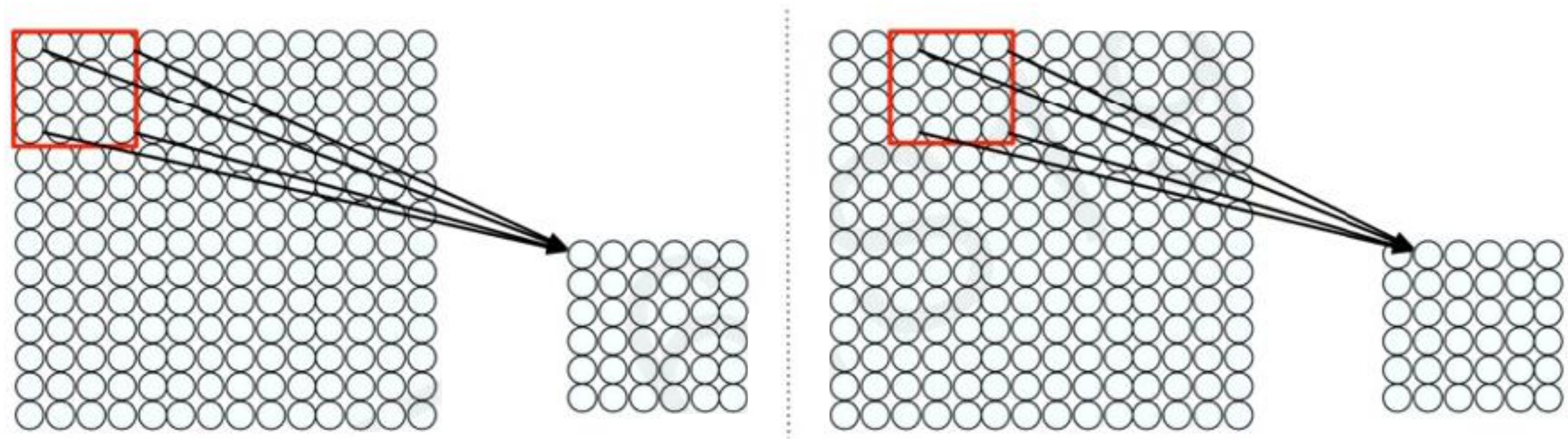
Connect patches of inputs to neurons in hidden layer

Concept

Only "looks" those values

How to connect each patch?

http://introtodeeplearning.com/



- Use a *sliding window* in input layer to a single neuron in subsequent layer.
- Add **weights**.
- **How can we** weight the batch to detect particular features?

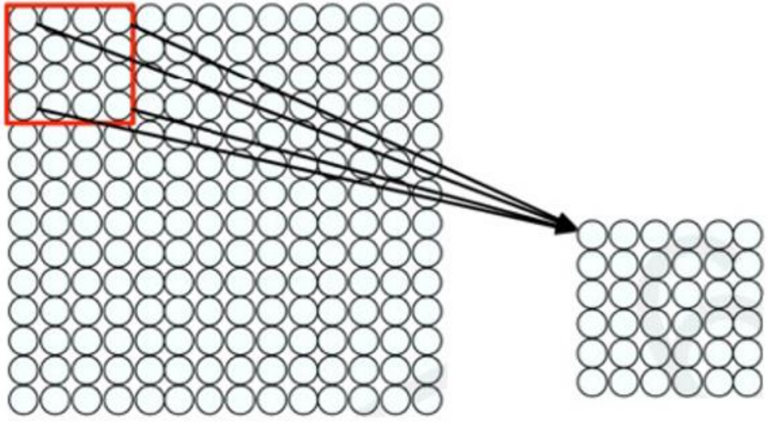
Applying Filter

http://introtodeeplearning.com/

- Apply a set of weights – a filter – to extract local features
- Use multiple filters to extract different features/
- Spatially share parameters of each filter

Convolution works!

http://introtodeeplearning.com/



- *“Patchy”* operation is convolution:
  - Filter of size 4x4 (16 weigths)
  - Apply this same filter to 4x4 patches in input
  - Shift by 2 pixels for next patch..

Feature Extraction and Convolution

X or X?

http://introtodeeplearning.com/

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	-1	-1	-1
-1	-1	1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	-1	-1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1

?

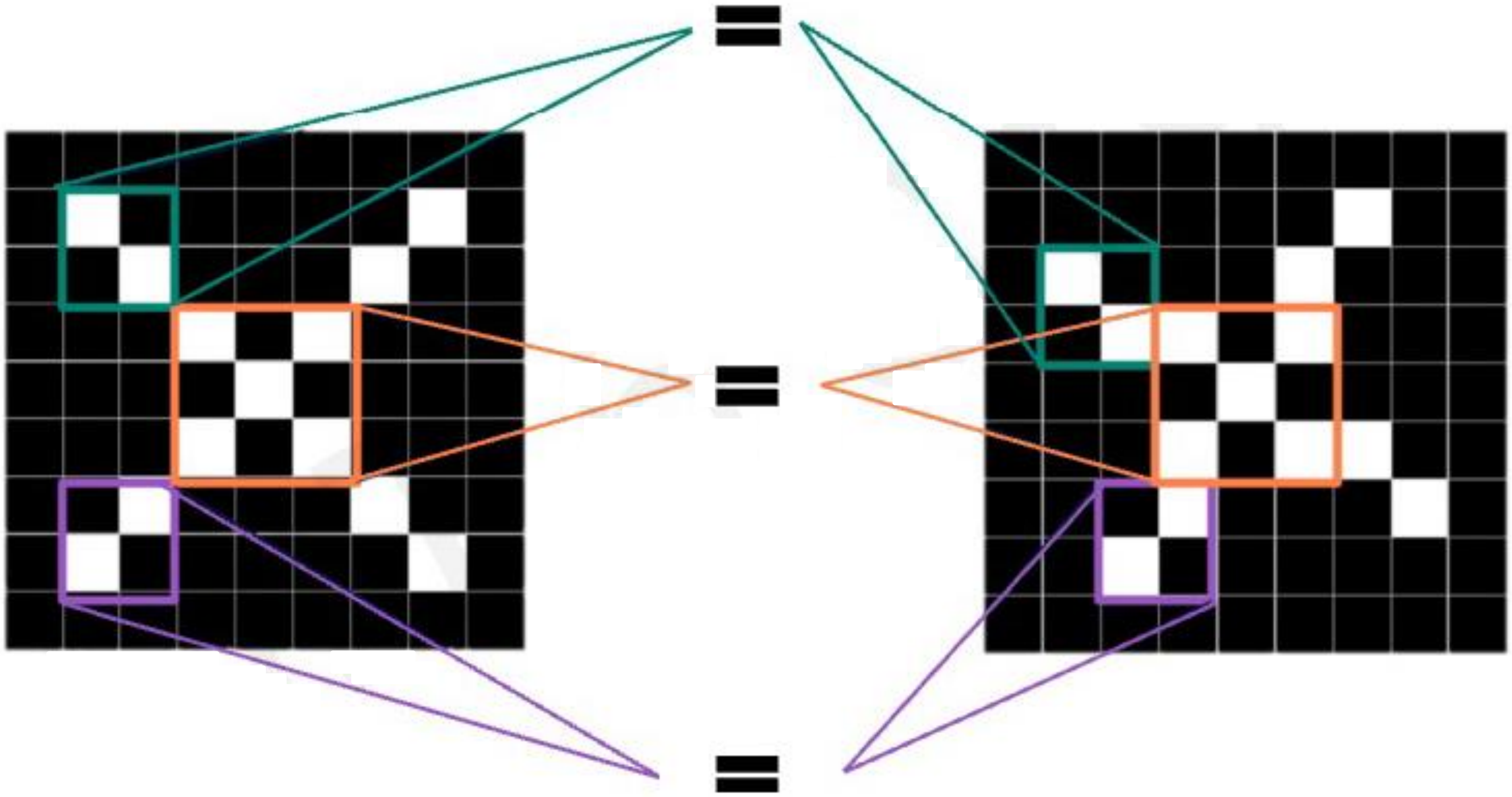
=

-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	-1	-1	-1	-1	-1	1	-1	-1
-1	1	-1	-1	-1	1	-1	-1	-1
-1	-1	1	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	1	-1	-1
-1	-1	-1	1	-1	-1	-1	1	-1
-1	-1	1	-1	-1	-1	-1	-1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1

We want to able to classify an X ad an X even if it's shifted, shrunk, rotated, deformed

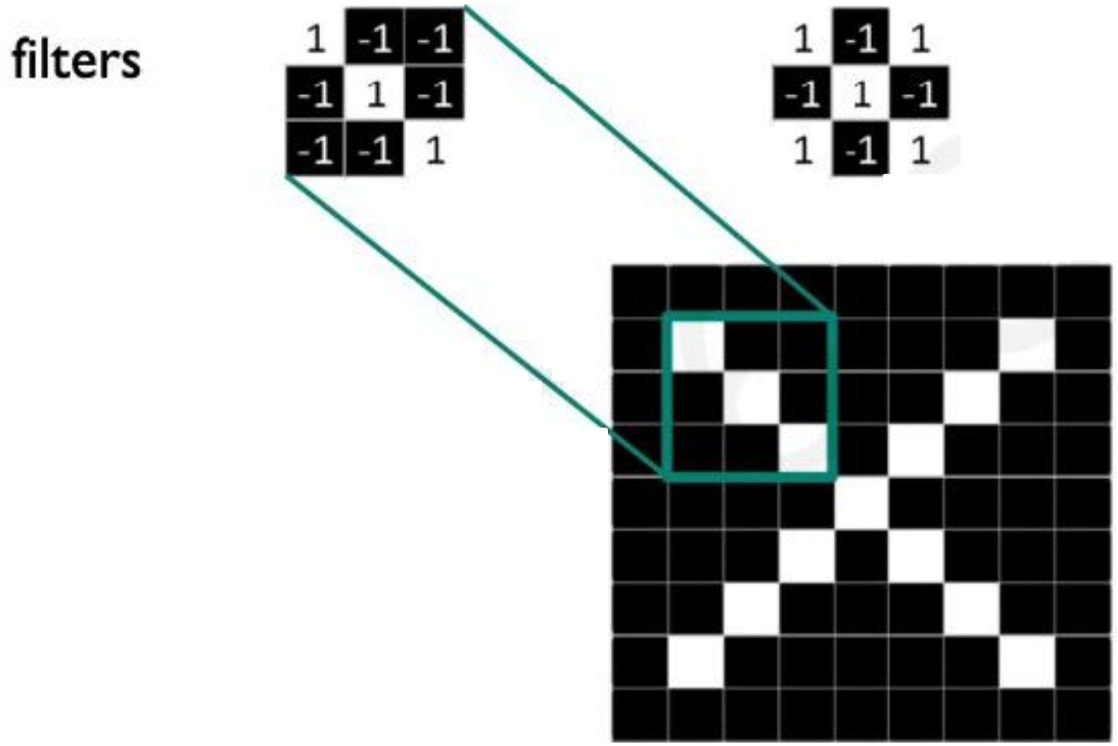
Features of X

http://intrc



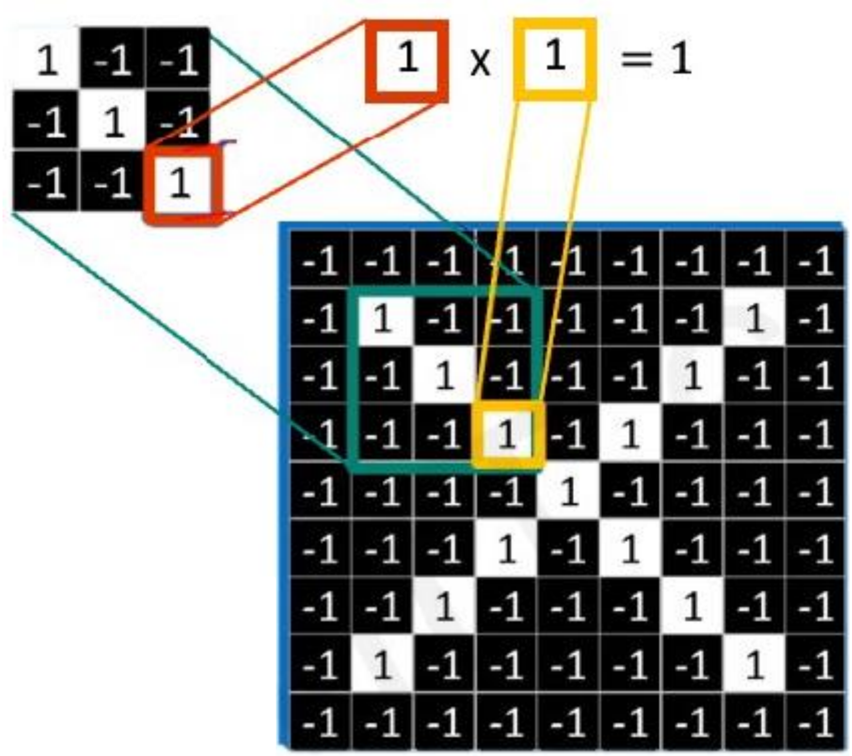
Filter to detect X features

http://introtodeeplearning.com/



Convolution Operation

http://introtodeeplearning.com/



Element-wise multiplication

Add outputs

$$\odot \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = 9$$

Example

http://introtodeeplearning.com/

Image

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

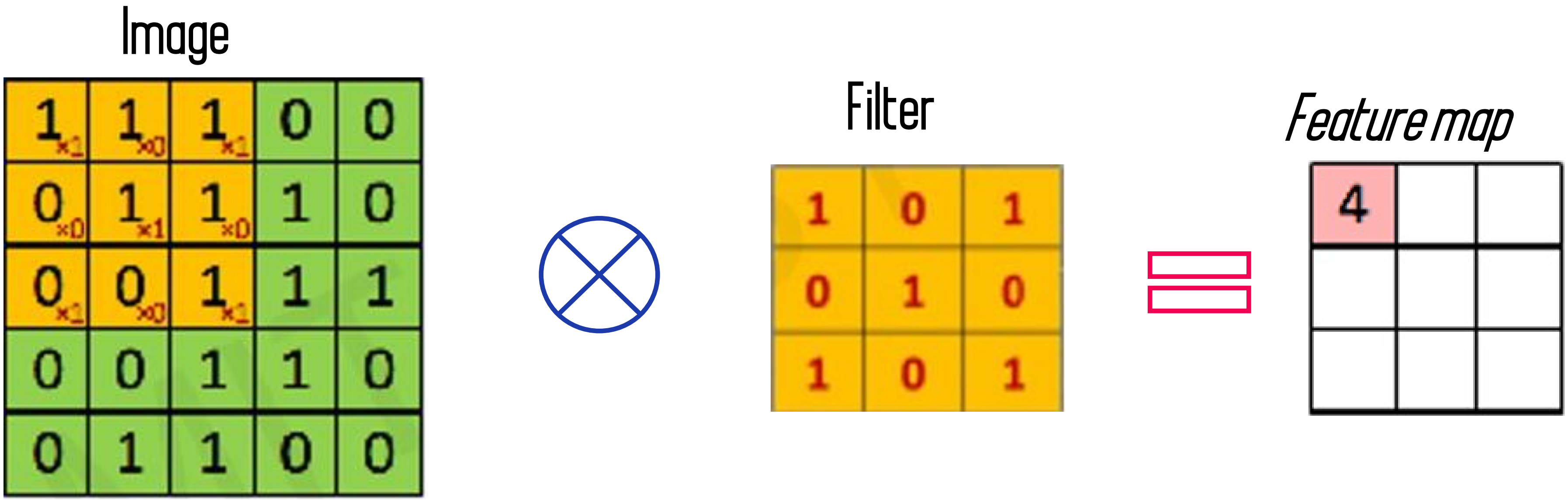
$$\otimes$$

Filter

1	0	1
0	1	0
1	0	1

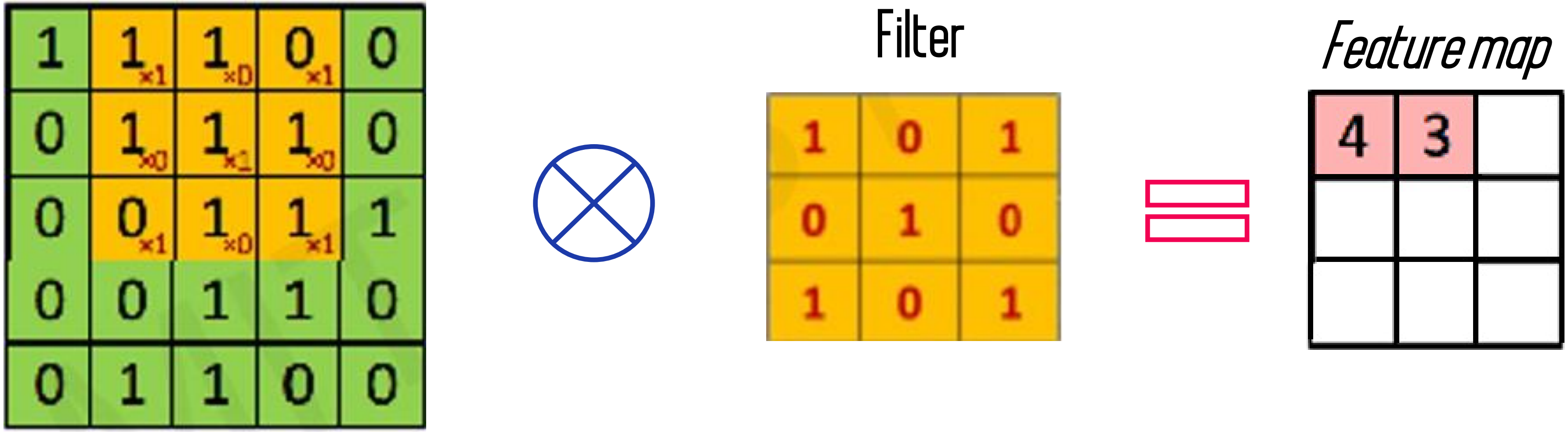
Example

http://introtodeeplearning.com/



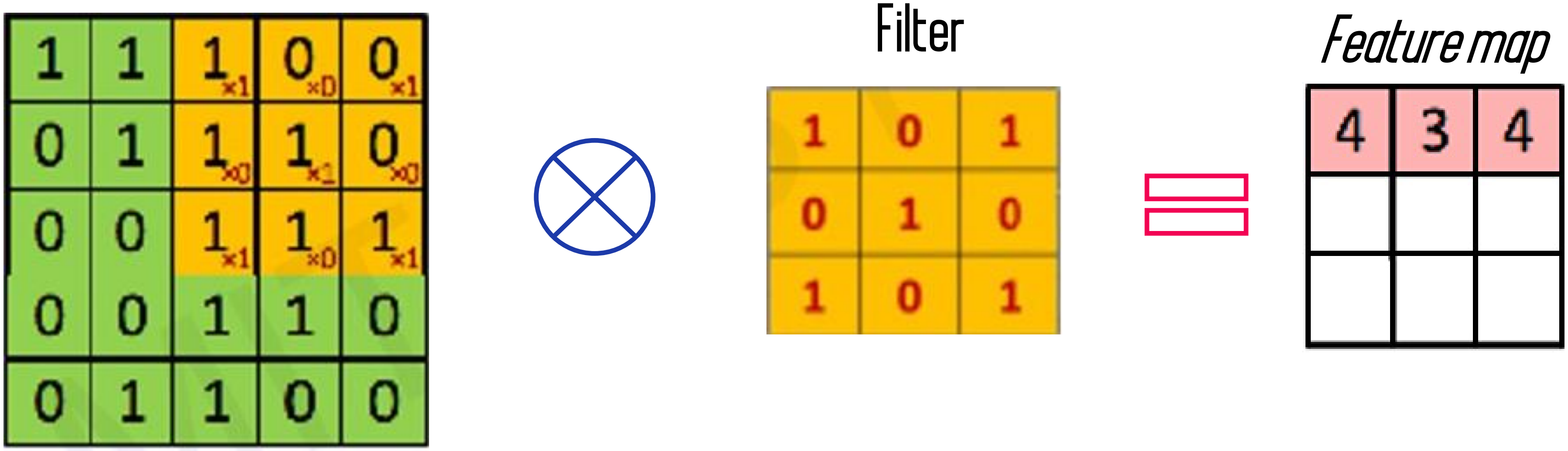
Example

http://introtodeeplearning.com/



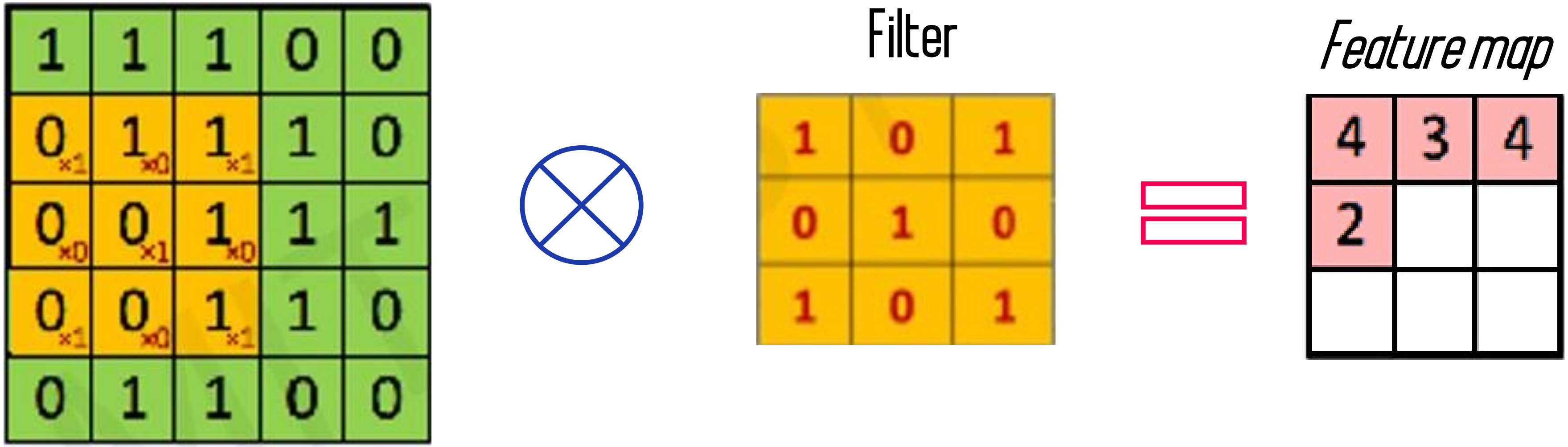
Example

http://introtodeeplearning.com/



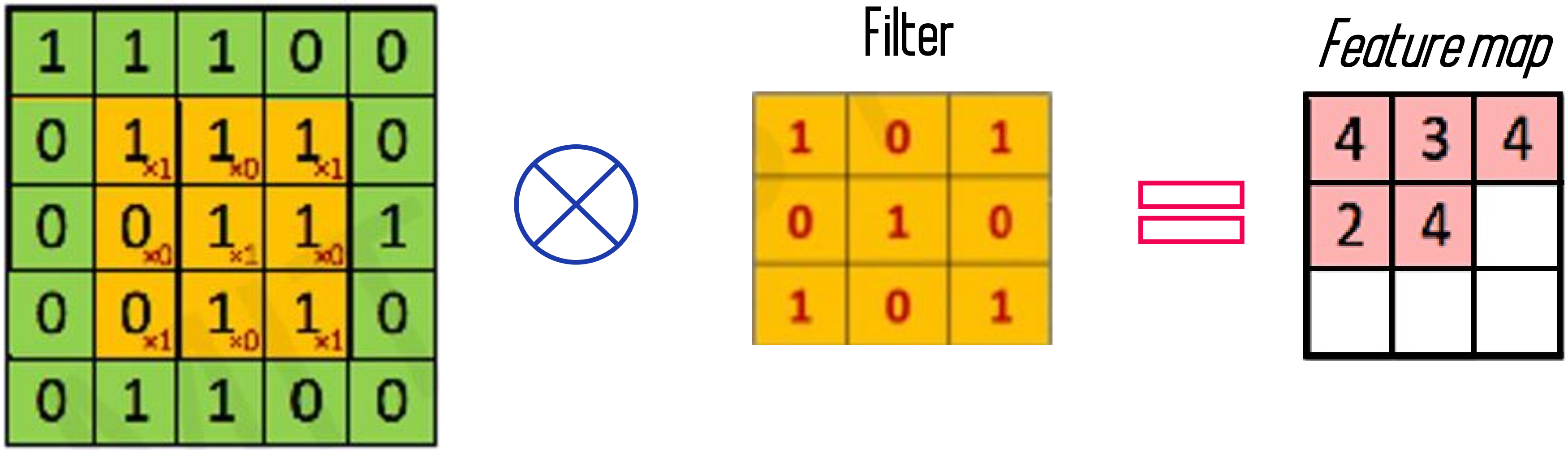
Example

http://introtodeeplearning.com/



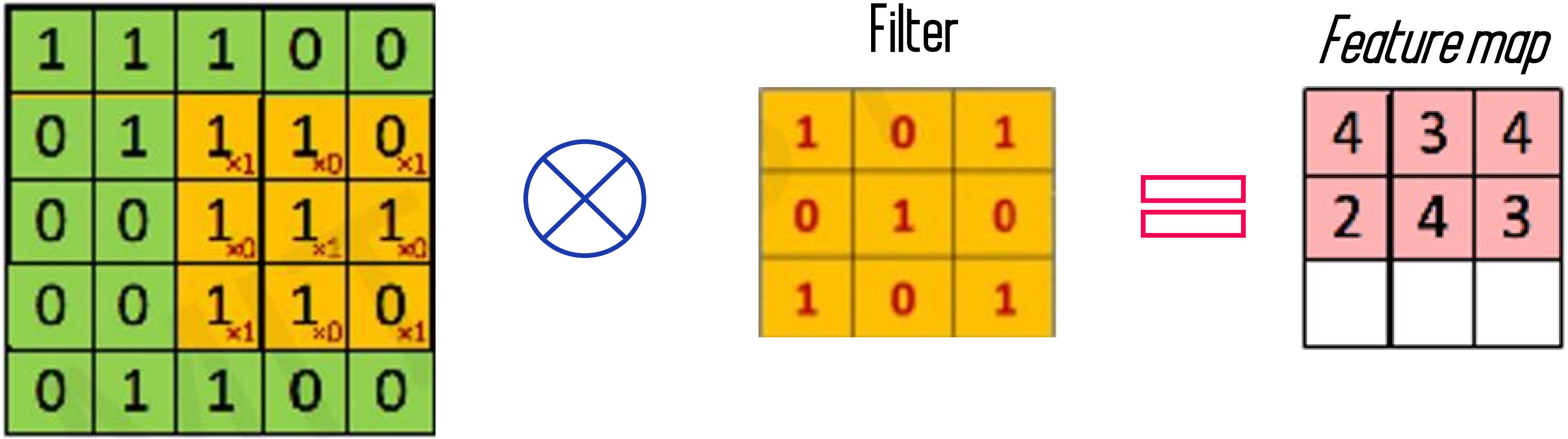
Example

http://introtodeeplearning.com/



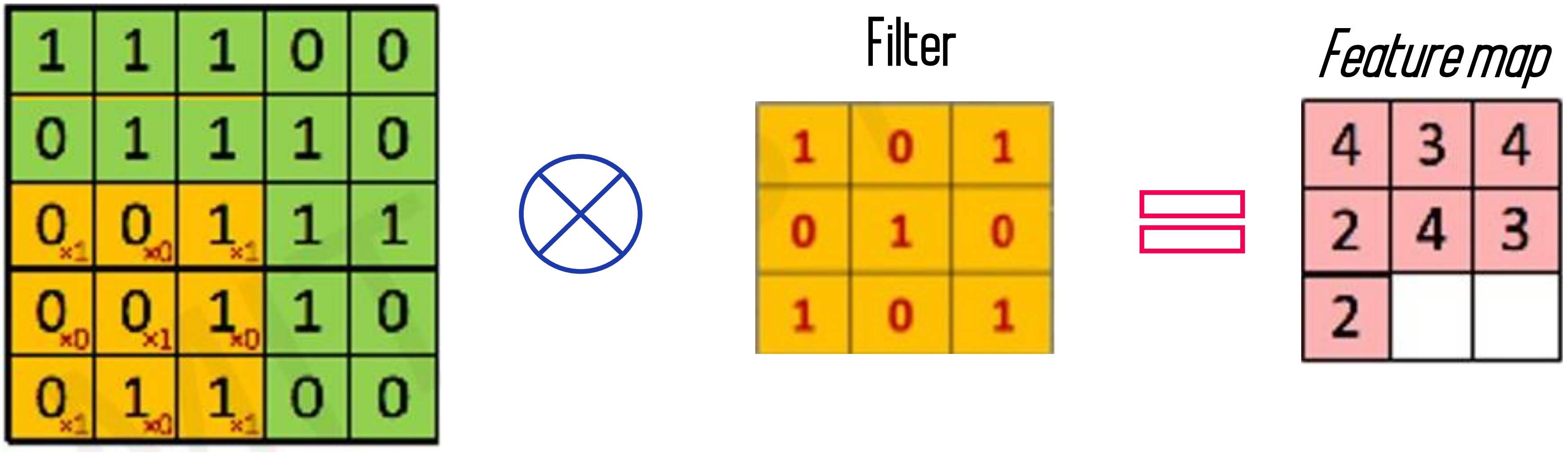
Example

http://introtodeeplearning.com/



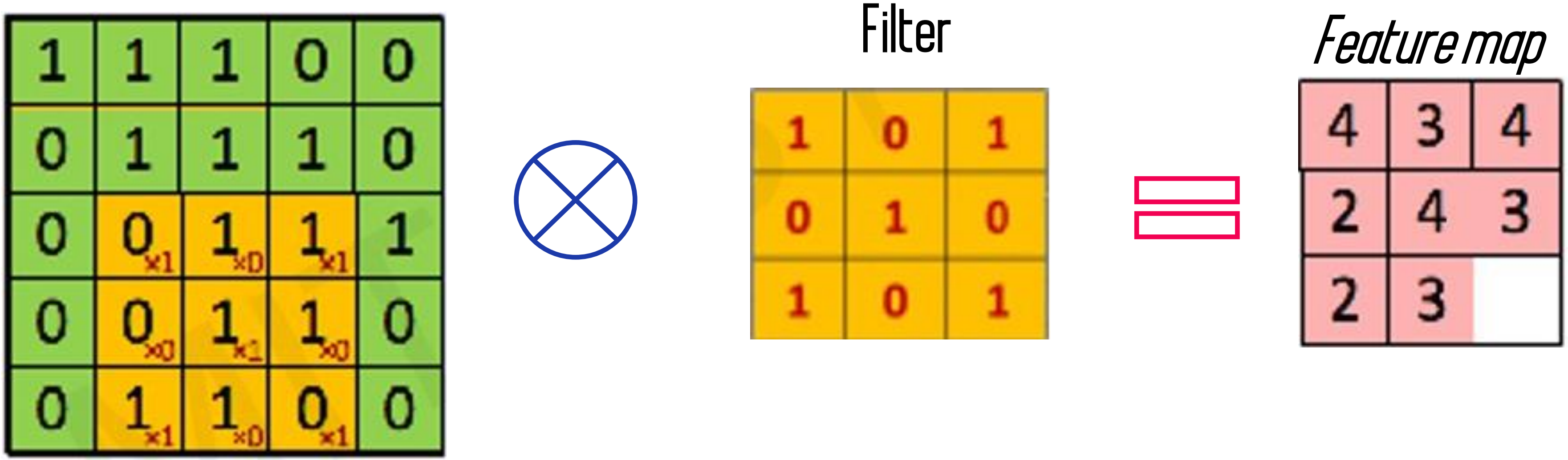
Example

http://introtodeeplearning.com/



Example

http://introtodeeplearning.com/



Feature Map Variation

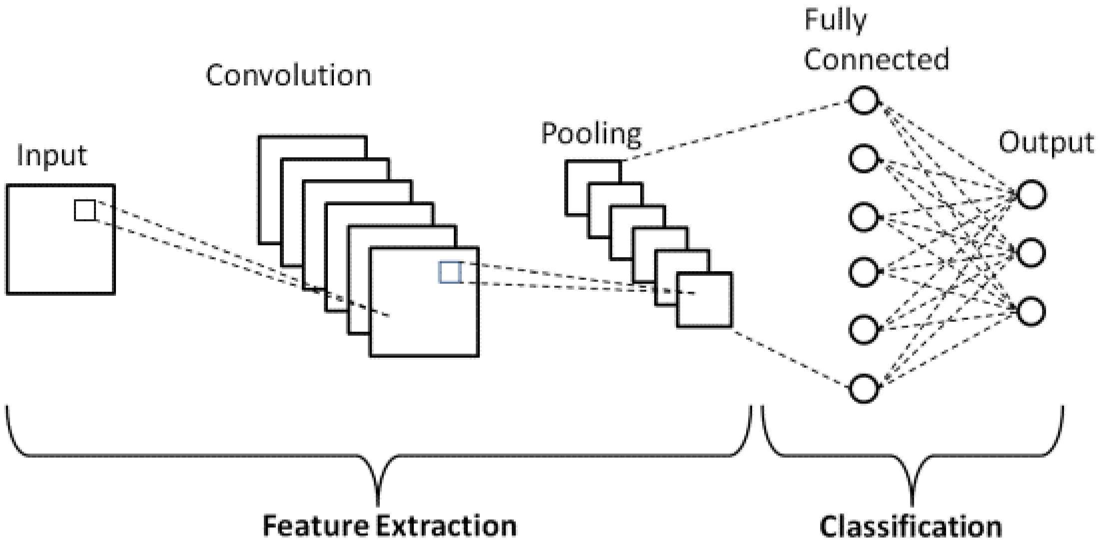
http://introtodeeplearning.com/

Original | Sharpen | Edge Detect | "Strong" Edge Detect

Convolutional Neural Networks

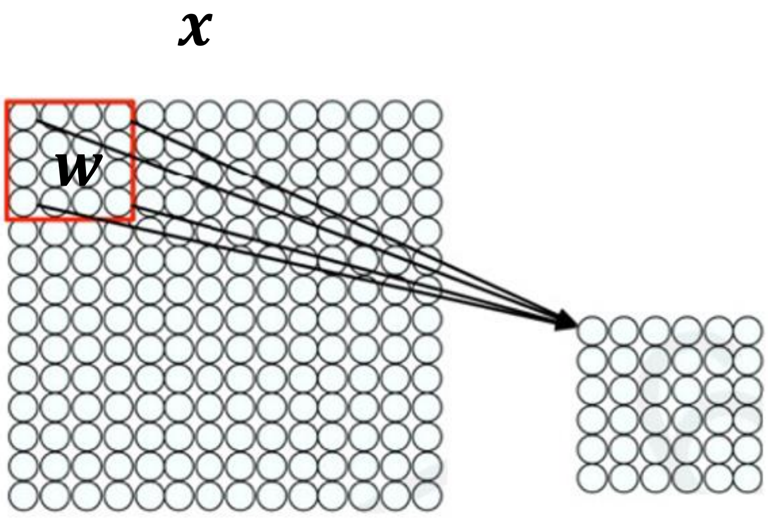
CNN for classification

http://introtodeeplearning.co



1. Convolution : apply filters to generate *feature map*
2. Non-linearitas : usually ReLU
3. Pooling : *downsampling* operation on each *feature map*.

```
tf.keras.layers.Conv2D
tf.keras.activations.*
tf.keras.layers.MaxPool2D
```



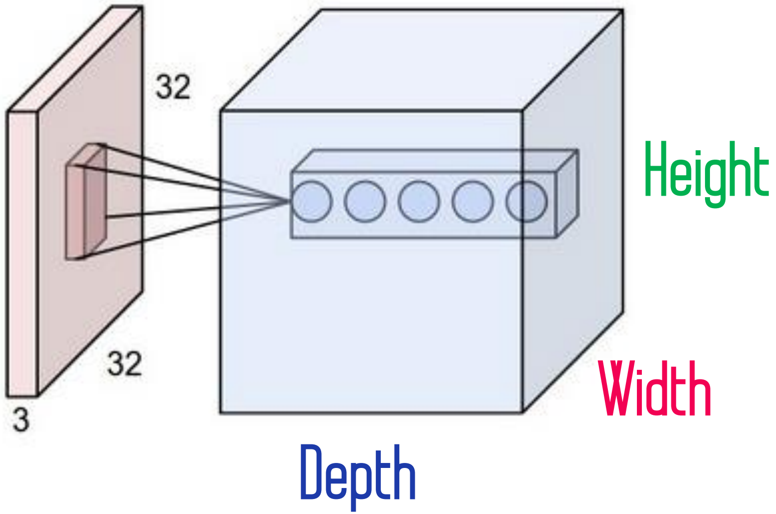
`tf.keras.layers.Conv2D`

$$\sum_{i=1}^{4}\sum_{i=1}^{4} w_{ij}x_{i+p,j+q} + b$$

For neuron (p, q) in hidden layer

Volume luaran terkait Spasial

http://introtodeeplearning.com/



Layer Dimension: h x w x d

h and w are spatial dimensions, d

Is number of filters

Stride: Filter step size

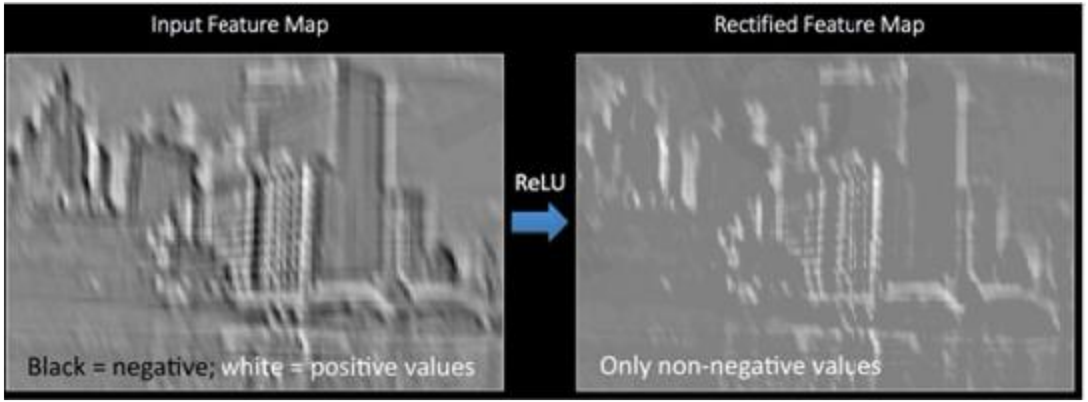
Receptive field: Location in input image that

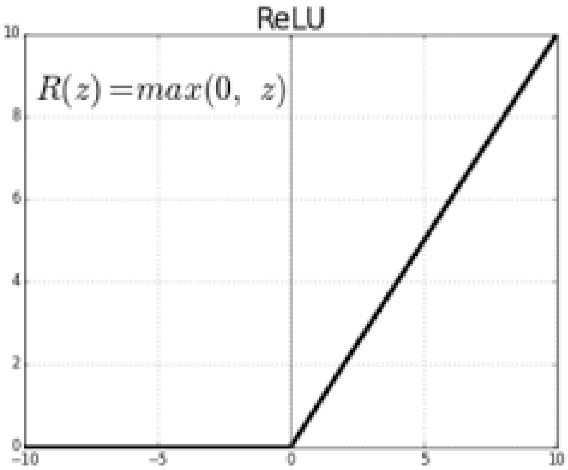
a node is path connected to

```
tf.keras.layers.Conv2D( filters=d, kernel_size=(h,w), strides=s )
```

Non-linearity and *Pooling*

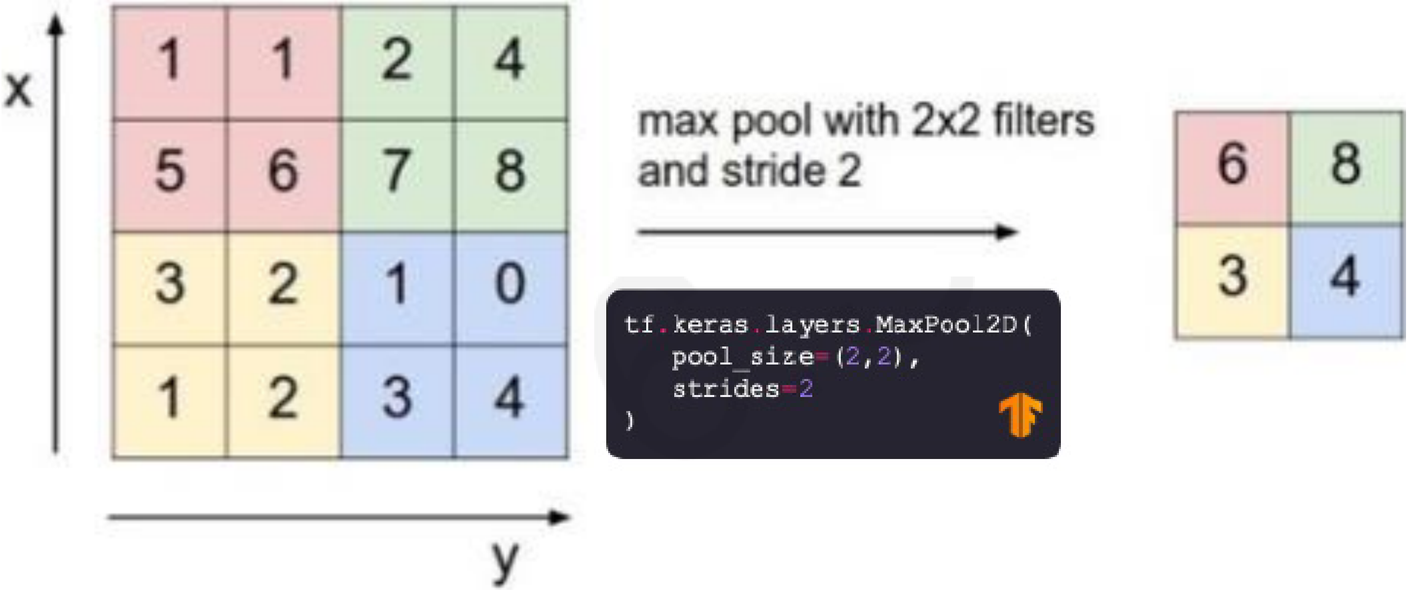
What is non-linearity?






```
tf.keras.layers.ReLU
```

What is Pooling?



- Reduce dimensionality
- *Spatial Invariance*

end-to-end code

Put all together

```
import tensorflow as tf

def generate_model():
    model = tf.keras.Sequential([
        # first convolutional layer
        tf.keras.layers.Conv2D(32, filter_size=3, activation='relu'),
        tf.keras.layers.MaxPool2D(pool_size=2, strides=2),

        # second convolutional layer
        tf.keras.layers.Conv2D(64, filter_size=3, activation='relu'),
        tf.keras.layers.MaxPool2D(pool_size=2, strides=2),

        # fully connected classifier
        tf.keras.layers.Flatten(),
        tf.keras.layers.Dense(1024, activation='relu'),
        tf.keras.layers.Dense(10, activation='softmax')   # 10 outputs
    ])
    return model
```

Discussion...

Q n A